

PREREQUISITES

The Significance of Human Communication

Communication is the process of exchanging information. People communicate to convey their thoughts, ideas, and feelings to others. The process of communication is inherent to all human life and includes verbal, nonverbal (body language), print, and electronic processes.

Two of the main barriers to human communication are language and distance.

Language barriers arise between persons of different cultures or nationalities.

Communicating over long distances is another problem. Communication between early human beings was limited to face-to-face encounters. Long-distance communication was first accomplished by sending simple signals such as drumbeats, horn blasts, and smoke signals and later by waving signal flags (semaphores). When messages were relayed from one location to another, even greater distances could be covered.

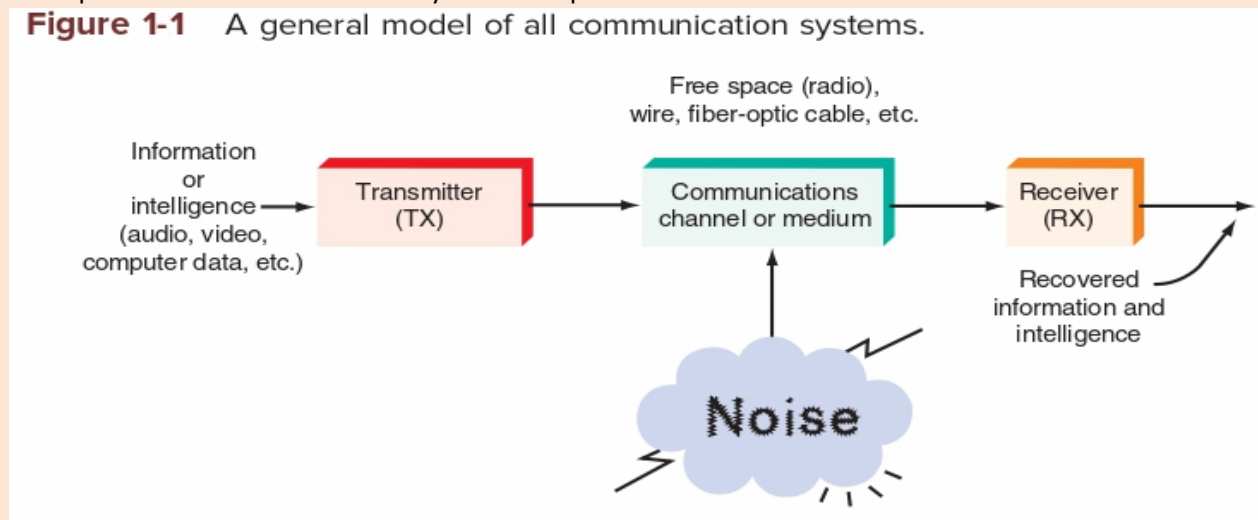
The distance over which communication could be sent was extended by the written word. For many years, long-distance communication was limited to the sending of verbal or written messages by human runner, horseback, ship, and later trains.

Well-known forms of electronic communication, such as the telephone, radio, TV, and the Internet, have increased our ability to share information. The way we do things and the success of our work and personal lives are directly related to how well we communicate. It has been said that the emphasis in our society has now shifted from that of manufacturing and mass production of goods to the accumulation, packaging, and exchange of information. Ours is an information society, and a key part of it is communication. Without electronic communication, we could not access and apply the available information in a timely way.

Communication Systems

All electronic communication systems have a transmitter, a communication channel or medium, and a receiver. These basic components are shown in Fig. 1-1. The process of communication begins when a human being generates some kind of message, data, or other intelligence that must be received by others.

A message may also be generated by a computer or electronic current. In *electronic communication systems*, the message is referred to as *information*, or an intelligence signal. This message, in the form of an electronic signal, is fed to the transmitter, which then transmits the message over the communication channel. The message is picked up by the receiver and relayed to another human. Along the way, noise is added in the communication channel and in the receiver. *Noise* is the general term applied to any phenomenon that degrades or interferes with the transmitted information.

Figure 1-1 A general model of all communication systems.

Transmitter

The *transmitter* itself is a collection of electronic components and circuits designed to convert the electrical signal to a signal suitable for transmission over a given communication medium. Transmitters are made up of oscillators, amplifiers, tuned circuits and filters, modulators, frequency mixers, frequency synthesizers, and other circuits. The original intelligence signal usually modulates a higher-frequency carrier sine wave generated by the transmitter, and the combination is raised in amplitude by power amplifiers, resulting in a signal that is compatible with the selected transmission medium.

Communication Channel

The *communication channel* is the medium by which the electronic signal is sent from one place to another. Many different types of media are used in communication systems, including wire conductors, fiber-optic cable, and free space.

- **Electrical Conductors.** In its simplest form, the medium may simply be a pair of wires that carry a voice signal from a microphone to a headset. It may be a coaxial cable such as that used to carry cable TV signals. Or it may be a twisted-pair cable used in a local-area network (LAN).
- **Optical Media.** The communication medium may also be a fiber-optic cable or “light pipe” that carries the message on a light wave. These are widely used today to carry long-distance calls and all Internet communications. The information is converted to digital form that can be used to turn a laser diode off and on at high speeds. Alternatively, audio or video analog signals can be used to vary the amplitude of the light.

- **Free Space.** When free space is the medium, the resulting system is known as radio. Also known as *wireless*, *radio* is the broad general term applied to any form of wireless communication from one point to another. Radio makes use of the electromagnetic spectrum. Intelligence signals are converted to electric and magnetic fields that propagate nearly instantaneously through space over long distances. Communication by visible or infrared light also occurs in free space.
- **Other Types of Media.** Although the most widely used media are conducting systems. For example, in sonar, water is used as the medium. Passive sonar “listens” for under- water sounds with sensitive hydrophones. Active sonar uses an echo-reflecting technique similar to that used in radar for determining how far away objects under water are and in what direction they are moving. The earth itself can be used as a communication medium, because it conducts electricity and can also carry low-frequency sound waves.

Receivers

A *receiver* is a collection of electronic components and circuits that accepts the transmitted message from the channel and converts it back to a form understandable by humans. Receivers contain amplifiers, oscillators, mixers, tuned circuits and filters, and a demodulator or detector that recovers the original intelligence signal from the modulated carrier. The output is the original signal, which is then read out or displayed. It may be a voice signal sent to a speaker, a video signal that is fed to an LCD screen for display, or binary data that is received by a computer and then printed out or displayed on a video monitor.

Transceivers

Most electronic communication is two-way, and so both parties must have both a transmitter and a receiver. As a result, most communication equipment incorporates circuits that both send and receive. These units are commonly referred to as *transceivers*. All the transmitter and receiver circuits are packaged within a single housing and usually share some common circuits such as the power supply. Telephones, handheld radios, cellular telephones, and computer modems are examples of transceivers.

Attenuation

Signal *attenuation*, or degradation, is inevitable no matter what the medium of transmission. Attenuation is proportional to the square of the distance between the transmitter and receiver.

Media are also frequency-selective, in that a given medium will act as a low-pass filter to a transmitted signal, distorting digital pulses in addition to greatly reducing signal amplitude over long distances. Thus considerable signal amplification, in both the transmitter and the receiver, is required for successful transmission. Any medium also slows signal propagation to a speed slower than the speed of light.

Noise

Noise is mentioned here because it is the bane of all electronic communications. Its effect is experienced in the receiver part of any communications system. While some noise can be filtered out, the general way to minimize noise is to use components that contribute less noise and to lower their temperatures. The measure of noise is usually expressed in terms of the signal-to-noise (S/N) ratio (SNR), which is the signal power divided by the noise power and can be stated numerically or in terms of decibels (dB). Obviously, a very high SNR is preferred for best performance

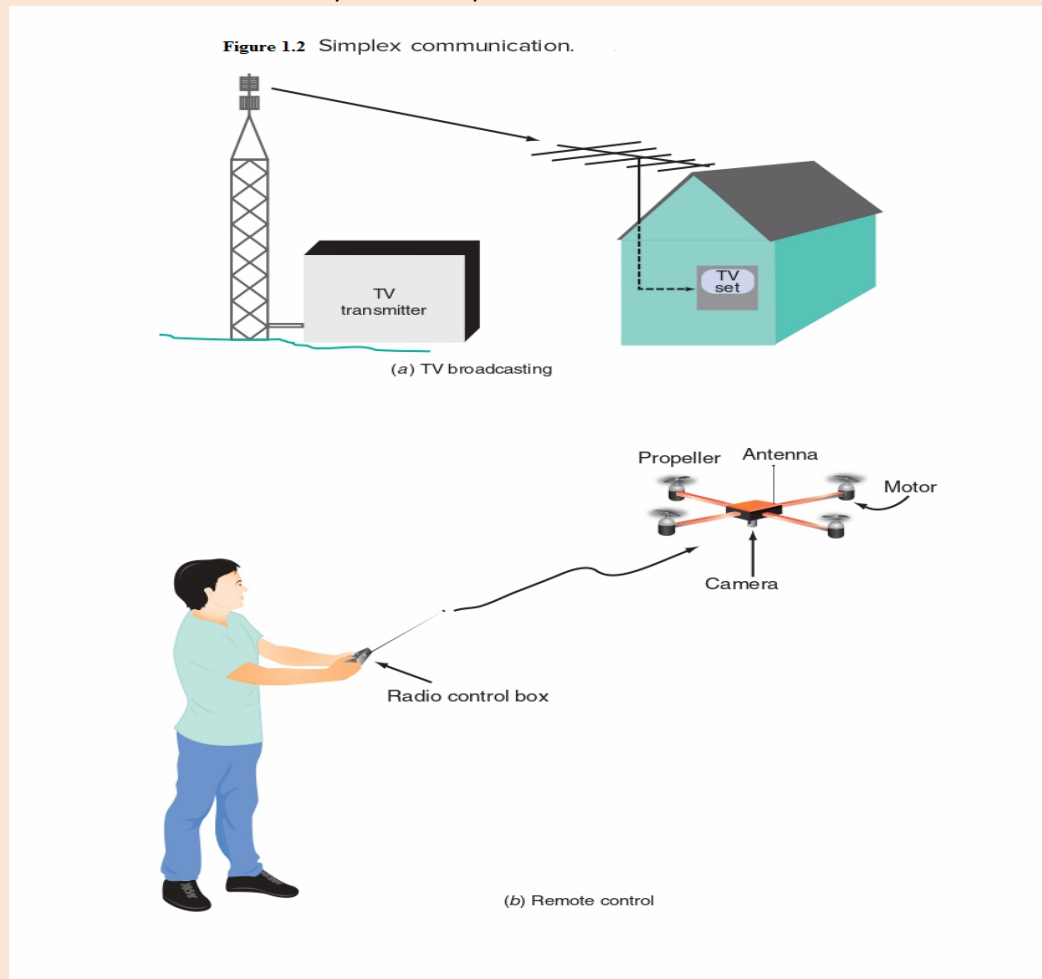
Types of Electronic Communication

Electronic communications are classified according to whether they are

one-way (simplex)
two-way (full duplex or half duplex) transmissions
analog or digital signals▪

• Simplex

The simplest way in which electronic communication is conducted is one-way communications, normally referred to as *simplex communication*. Examples are shown in Fig. 1-2. The most common forms of simplex communication are radio and TV broadcasting. Another example of one-way communication is transmission to a remotely controlled vehicle like a toy car or an unmanned aerial vehicle (UAV or drone)



• Full Duplex

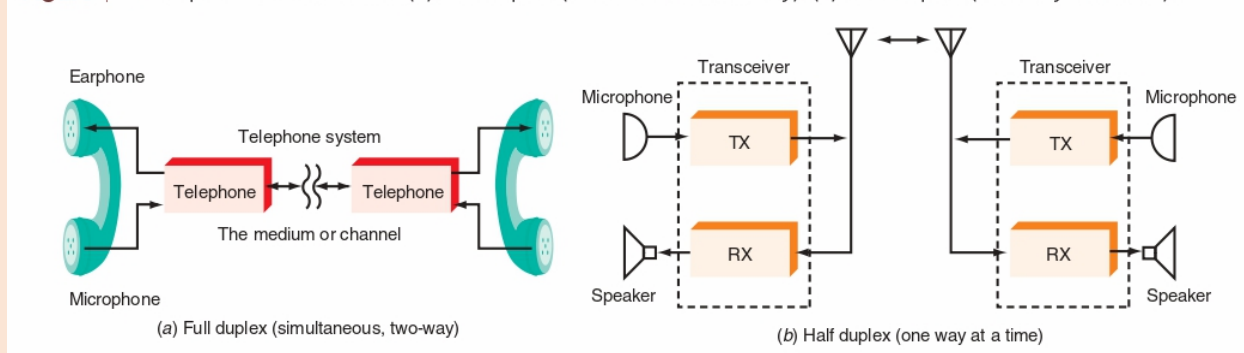
The bulk of electronic communication is two-way, or *duplex communication*. Typical duplex applications are shown in Fig. 1-3.

For example, people communicating with one another over the telephone can talk and listen simultaneously, as Fig. 1-3(a) illustrates. This is called *full duplex communication*.

• Half Duplex

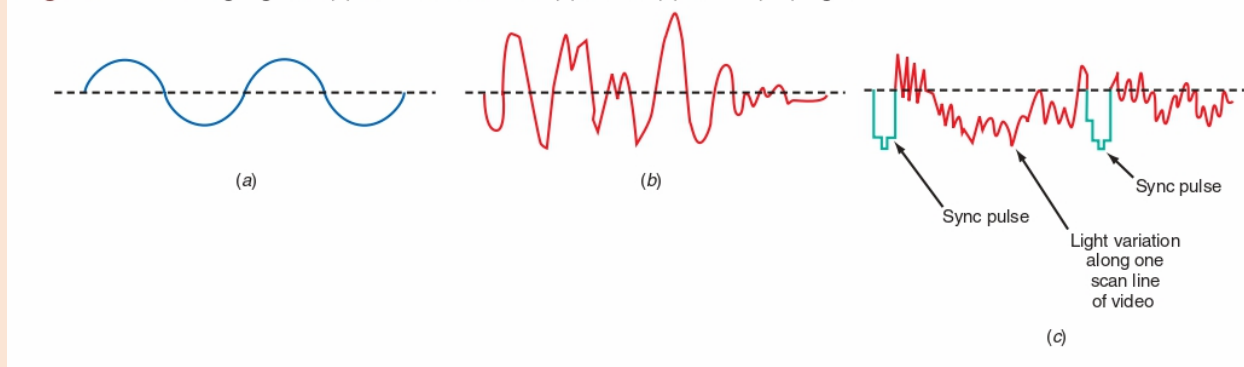
The form of two-way communication in which only one party transmits at a time is known as *half duplex communication* [see Fig. 1-3(b)].

The communication is two-way, but the direction alternates: the communicating parties take turns transmitting and receiving. Most radio transmissions, such as those used in the military, fire, police, aircraft, marine, and other services, are half duplex communication. Citizens band (CB), Family Radio, and amateur radio communication are also half duplex.

Figure 1.3 Duplex communication. (a) Full duplex (simultaneous two-way). (b) Half duplex (one way at a time).

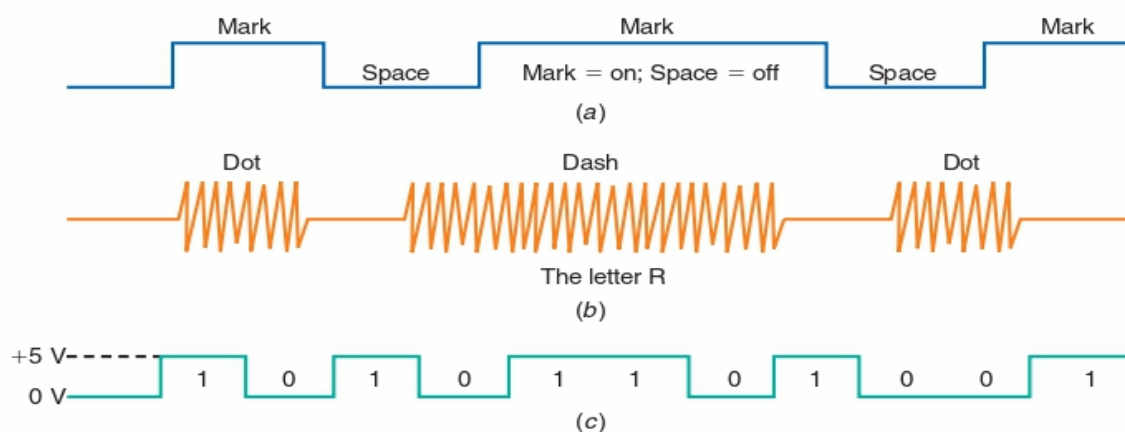
Analog Signals

An *analog signal* is a smoothly and continuously varying voltage or current. Some typical analog signals are shown in Fig. 1-4. A sine wave is a single-frequency analog signal. Voice and video voltages are analog signals that vary in accordance with the sound or light variations that are analogous to the information being transmitted.

Figure 1.4 Analog signals. (a) Sine wave “tone.” (b) Voice. (c) Video (TV) signal.

Digital Signals

Digital signals, in contrast to analog signals, do not vary continuously, but change in steps or in discrete increments. Most digital signals use binary or two-state codes.

Figure 1.5 Digital signals. (a) Telegraph (Morse code). (b) Continuous-wave (CW) code. (c) Serial binary code.

The earliest forms of both wire and radio communication used a type of on /off digital code. The

telegraph used Morse code, with its system of short and long signals (dots and dashes) to designate letters and numbers. See Fig. 1-5(a). In radio telegraphy, also known as continuous-wave (CW) transmission, a sine wave signal is turned off and on for short or long durations to represent the dots and dashes. Refer to Fig. 1-5(b).

Data used in computers is also digital. Binary codes representing numbers, letters, and special symbols are transmitted serially by wire, radio, or optical medium. The most commonly used digital code in communications is the *American Standard Code for Information Interchange* (ASCII, pronounced “ask key”). Fig. 1-5(c) shows a serial binary code

Modulation and Multiplexing

Modulation and multiplexing are electronic techniques for transmitting information efficiently from one place to another.

Modulation makes the information signal more compatible with the medium

Multiplexing allows more than one signal to be transmitted concurrently over a single medium. Modulation and multiplexing techniques are basic to electronic communication

Baseband Transmission

Before it can be transmitted, the information or intelligence must be converted to an electronic signal compatible with the medium.

For example, a microphone changes voice signals (sound waves) into an analog voltage of varying frequency and amplitude. This signal is then passed over wires to a speaker or headphones. This is the way the telephone system works.

A video camera generates an analog signal that represents the light variations along one scan line of the picture. This analog signal is usually transmitted over a coaxial cable. Binary data is generated by a keyboard attached to a computer

The computer stores the data and processes it in some way. The data is then transmitted on cables to peripherals such as a printer or to other computers over a LAN.

Regardless of whether the original information or intelligence signals are analog or digital, they are all referred to as baseband signals

In a communication system, baseband information signals can be sent directly and unmodified over the medium or can be used to modulate a carrier for transmission over the medium. Putting the original voice, video, or digital signals directly into the medium is referred to as *baseband transmission*. For example, in many telephone and intercom systems, it is the voice itself that is placed on the wires and transmitted over some distance to the receiver. In most computer networks, the digital signals are applied directly to coaxial or twisted-pair cables for transmission to another computer.

In many instances, baseband signals are incompatible with the medium. Although it is theoretically possible to transmit voice signals directly by radio, realistically it is impractical. As a result, the baseband information signal, be it audio, video, or data, is normally used to modulate a high-frequency signal called a *carrier*. The higher-frequency carriers radiate into space more

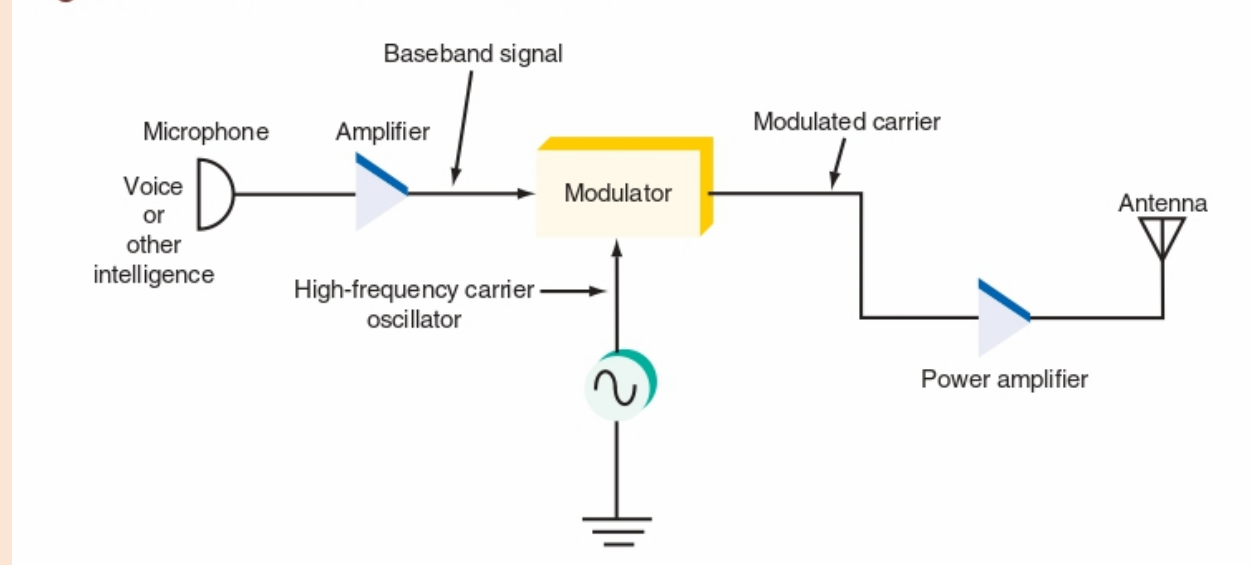
efficiently than the baseband signals themselves. Such wireless signals consist of both electric and magnetic fields. These electromagnetic signals, which are able to travel through space for long distances, are also referred to as *radio-frequency (RF) waves*, or just radio waves.

Broadband Transmission

Modulation is the process of having a baseband voice, video, or digital signal modify another, higher-frequency signal, the carrier. The process is illustrated in Fig. 1-6.

The information or intelligence to be sent is said to be *impressed* upon the carrier. The carrier is usually a sine wave generated by an oscillator. The carrier is fed to a circuit called a modulator along with the baseband intelligence signal. The intelligence signal changes the carrier in a unique way. The modulated carrier is amplified and sent to the antenna for transmission. This process is called *broadband transmission*.

Figure 1.6 Modulation at the transmitter.



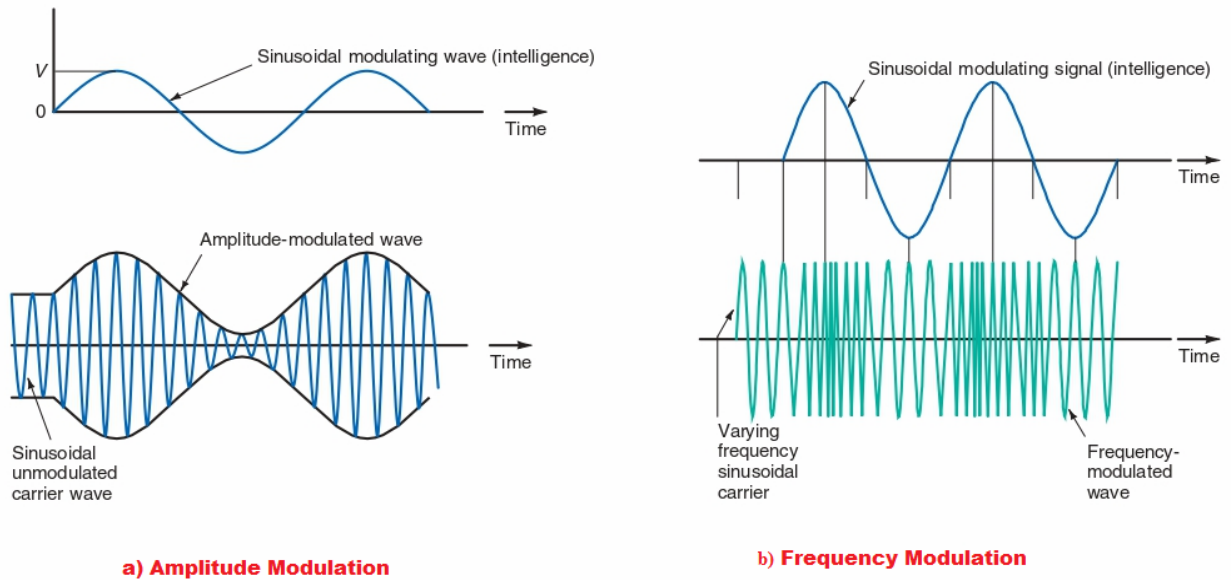
The three ways to make the baseband signal change the carrier sine wave are to vary its amplitude, vary its frequency, or vary its phase angle.

The two most common methods of modulation are *amplitude modulation (AM)* and *frequency modulation (FM)*.

In AM, the baseband information signal called the modulating signal varies the amplitude of the higher-frequency carrier signal, as shown in Fig. 1-7(a).

In FM, the information signal varies the frequency of the carrier, as shown in Fig. 1-7(b)

Figure 1.7 Types of modulation. (a) Amplitude modulation. (b) Frequency modulation.



Varying the phase angle produces *phase modulation (PM)*.